Appendix II. Debt Sustainability Analysis

An analysis of the impact of the Ebola outbreak on debt sustainability indicates that the risk of debt distress remains low. The external and public debt profiles are somewhat less favorable than in the July 2014 DSA update, mostly owing to lower growth in 2014–15. In particular, the average external debt to GDP ratio in 2015–34 is projected to be 29.9 percent, only marginally higher (by 0.5 percent) than at the time of the July 2014 update. However, a number of factors mitigate the impact of the crisis on the external and public debt paths (Appendix Tables 1 and 2).

- First, the economy is assumed to recover starting in 2016 as the shock is of a temporary nature (albeit protracted). Average annual real GDP growth in 2014–16 has been revised downward, from 7 to 2 percent. Nonetheless, medium-term growth is also affected by delays in infrastructure and mining projects.
- Second, external loan disbursements in FY2014 (pre-Ebola crisis) have been slow compared with staff's previous projections. Total external loan disbursements between 2014 and 2033 are now projected to be about US$0.5 billion below the amount projected at the time of the last DSA update.
- Third, the projected large financing gaps related to the Ebola epidemic are assumed to be filled mostly with grants and highly concessional loans, and some planned project loan disbursements have been delayed.

Stress tests also indicate that external and public debt would remain sustainable over the projection period. Owing to the relatively low level of debt outstanding at present, and the highly concessional terms of the new borrowing, the external and public debt profiles would remain sustainable (Appendix Figures 1 and 2).

Appendix Table 1. Liberia: External Debt Sustainability Framework, Baseline Scenario, 2011–34[1]

(Percent of GDP, unless otherwise indicated)

	Actual			Historical[7/] Average	Standard[7/] Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
External debt (nominal) 1/	**10.4**	**9.4**	**9.9**			**12.7**	**23.9**	**28.7**	**32.6**	**34.0**	**34.5**		**32.2**	**27.1**	
of which: public and publicly guaranteed (PPG)	10.4	9.4	9.9			12.7	23.9	28.7	32.6	34.0	34.5		32.2	27.1	
Change in external debt	2.5	-1.0	0.5			2.8	11.2	4.8	3.8	1.4	0.4		-0.9	0.0	
Identified net debt-creating flows	1.7	0.0	0.9			6.9	16.4	11.5	2.5	0.6	-0.4		-1.2	1.0	
Non-interest current account deficit	**29.5**	**24.2**	**24.9**	**20.6**	**11.8**	**30.0**	**35.9**	**33.6**	**27.6**	**29.8**	**32.5**		**13.3**	**10.6**	14.7
Deficit in balance of goods and services	88.8	77.3	65.3			64.1	68.5	61.5	49.9	45.5	44.0		19.6	16.9	
Exports	44.4	48.3	48.4			41.4	33.1	30.2	29.0	27.5	27.0		36.1	28.1	
Imports	133.2	125.6	113.7			105.5	101.7	91.7	78.9	73.0	71.0		55.7	44.9	
Net current transfers (negative = inflow)	-73.0	-66.5	-57.4	-102.1	36.2	-51.7	-48.6	-42.9	-37.2	-31.9	-27.2		-16.3	-10.8	-15.0
of which: official	-28.4	-26.7	-25.0			-25.4	-26.3	-24.9	-22.7	-20.8	-18.8		-13.1	-8.7	
Other current account flows (negative = net inflow)	13.6	13.4	17.0			17.6	16.0	14.9	14.8	16.2	15.7		10.0	4.6	
Net FDI (negative = inflow) 2/	**-26.7**	**-22.8**	**-23.1**	**-11.9**	**17.4**	**-22.8**	**-19.7**	**-21.8**	**-23.5**	**-26.9**	**-30.3**		**-13.0**	**-9.9**	-14.1
Endogenous debt dynamics 3/	**-1.1**	**-1.4**	**-0.9**			**-0.3**	**0.2**	**-0.2**	**-1.5**	**-2.2**	**-2.5**		**-1.5**	**0.3**	
Contribution from nominal interest rate	0.0	0.0	0.1			0.1	0.1	0.2	0.2	0.3	0.3		0.3	0.3	
Contribution from real GDP growth	-0.5	-0.7	-0.7			-0.4	0.1	-0.4	-1.7	-2.5	-2.8		-1.9	0.0	
Contribution from price and exchange rate changes	-0.7	-0.7	-0.4			...	...	...	...	...	...		...	...	
Residual (3-4) 4/	**0.8**	**-1.0**	**-0.4**			**-4.1**	**-5.2**	**-6.7**	**1.3**	**0.8**	**0.8**		**0.3**	**-1.0**	
of which: exceptional financing	0.0	0.0	0.0			0.0	-0.1	-0.3	-0.4	-0.2	0.0		0.0	0.0	
PV of external debt 5/	...	...	5.9			7.4	13.9	16.4	18.7	20.3	20.9		19.0	16.2	
In percent of exports	...	...	12.1			17.9	42.0	54.2	64.3	73.6	77.4		52.7	57.7	
PV of PPG external debt	**...**	**...**	**5.9**			**7.4**	**13.9**	**16.4**	**18.7**	**20.3**	**20.9**		**19.0**	**16.2**	
In percent of exports	**...**	**...**	**12.1**			**17.9**	**42.0**	**54.2**	**64.3**	**73.6**	**77.4**		**52.7**	**57.7**	
In percent of government revenues	**...**	**...**	**21.4**			**31.4**	**66.5**	**74.9**	**83.5**	**91.1**	**93.1**		**88.2**	**64.3**	
Debt service-to-exports ratio (in percent)	**0.7**	**0.1**	**0.8**			**0.6**	**1.0**	**1.6**	**1.3**	**1.5**	**3.4**		**3.6**	**3.9**	
PPG debt service-to-exports ratio (in percent)	**0.7**	**0.1**	**0.8**			**0.6**	**1.0**	**1.6**	**1.3**	**1.5**	**3.4**		**3.6**	**3.9**	
PPG debt service-to-revenue ratio (in percent)	**1.3**	**0.2**	**1.4**			**1.0**	**1.6**	**2.3**	**1.7**	**1.8**	**4.1**		**6.0**	**4.4**	
Total gross financing need (Millions of U.S. dollars)	43.9	24.2	41.4			148.9	337.7	262.4	103.7	85.8	91.8		79.8	185.5	
Non-interest current account deficit that stabilizes debt ratio	27.0	25.3	24.4			27.2	24.8	28.7	23.7	28.3	32.1		14.2	10.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	6.8	7.9	8.5	7.3	1.9	4.4	-0.4	1.8	6.4	8.5	9.4	5.0	6.1	-0.1	4.9
GDP deflator in US dollar terms (change in percent)	9.0	7.6	4.0	6.7	2.4	3.1	2.8	3.0	3.2	3.2	3.2	3.1	3.2	5.2	3.4
Effective interest rate (percent) 6/	0.4	0.1	1.6	0.5	0.7	0.8	0.9	0.7	0.8	0.9	1.0	0.8	1.1	1.1	1.1
Growth of exports of G&S (US dollar terms, in percent)	25.5	26.1	13.1	17.4	16.5	-7.9	-18.2	-4.4	5.6	6.1	10.9	-1.3	8.9	5.6	8.9
Growth of imports of G&S (US dollar terms, in percent)	14.6	9.3	2.2	16.4	25.3	-0.2	-1.3	-5.4	-5.5	3.6	9.8	0.2	6.2	5.0	5.2
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	47.9	43.1	53.2	48.4	38.1	39.2	45.0	48.1	48.1	48.1
Government revenues (excluding grants, in percent of GDP)	23.6	26.2	27.5			23.6	20.9	21.9	22.4	22.2	22.4		21.6	25.2	23.5
Aid flows (in Millions of US dollars) 8/	40.3	28.3	45.7			74.8	173.3	56.8	47.1	52.7	59.5		98.0	200.1	
of which: Grants	40.3	28.3	45.7			74.8	173.3	56.8	47.1	52.7	59.5		98.0	200.1	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 9/	...	...	...			5.5	13.6	5.8	5.1	3.9	4.0		3.4	3.0	3.3
Grant-equivalent financing (in percent of external financing) 9/	...	...	...			74.5	66.9	67.6	60.8	55.8	56.6		69.3	73.1	70.5
Memorandum items:															
Nominal GDP (Millions of US dollars)	1416	1643	1854			1995	2042	2142	2353	2634	2975		4899	10006	
Nominal dollar GDP growth	16.4	16.0	12.8			7.6	2.4	4.9	9.9	12.0	13.0	8.3	9.5	5.1	8.4
PV of PPG external debt (in Millions of US dollars)			105.3			143.4	281.1	344.6	430.0	522.3	611.2		917.3	1596.8	
(PVt-PVt-1)/GDPt-1 (in percent)						2.1	6.9	3.1	4.0	3.9	3.4	3.9	1.2	0.8	1.2
Gross workers' remittances (Millions of US dollars)	...	...	...			...	...	...	...	...	...		...	...	
PV of PPG external debt (in percent of GDP + remittances)	...	...	5.9			7.4	13.9	16.4	18.7	20.3	20.9		19.0	16.2	
PV of PPG external debt (in percent of exports + remittances)	...	...	12.1			17.9	42.0	54.2	64.3	73.6	77.4		52.7	57.7	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	0.8			0.6	1.0	1.6	1.3	1.5	3.4		3.6	3.9	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Includes private financing flows, including for iron-ore related investment which was included in FDI in the previous DSA

3/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

4/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

5/ Assumes that PV of private sector debt is equivalent to its face value.

6/ Current-year interest payments divided by previous period debt stock.

7/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

8/ Defined as grants, concessional loans, and debt relief.

9/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Appendix Table 2. Liberia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–34

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	12.3	11.0	11.4			15.3	25.1	29.3	33.0	34.8	35.3		33.0	27.9	
of which: foreign-currency denominated	12.3	11.0	11.4			15.3	25.1	29.3	32.6	34.1	34.5		32.2	27.1	
Change in public sector debt	2.0	-1.3	0.4			3.9	9.8	4.3	3.7	1.8	0.4		-0.9	0.0	
Identified debt-creating flows	-113.2	1.5	0.6			0.3	4.1	8.5	3.5	1.2	0.4		-1.4	-1.1	
Primary deficit	0.5	3.2	1.4	-0.4	2.0	1.0	4.6	9.2	5.8	4.4	4.1	4.9	1.2	-0.1	0.6
Revenue and grants	26.5	27.9	29.9			27.3	29.4	24.5	24.4	24.2	24.4		23.6	27.2	
of which: grants	2.8	1.7	2.5			3.8	8.5	2.7	2.0	2.0	2.0		2.0	2.0	
Primary (noninterest) expenditure	27.0	31.2	31.4			28.3	34.0	33.8	30.2	28.7	28.6		24.7	27.1	
Automatic debt dynamics	-1.6	-1.7	-0.8			-0.7	-0.4	-0.7	-2.3	-3.3	-3.8		-2.6	-1.0	
Contribution from interest rate/growth differential	-0.8	-1.1	-0.9			-0.5	0.0	-0.6	-2.1	-3.0	-3.4		-2.2	-0.2	
of which: contribution from average real interest rate	-0.1	-0.2	0.0			0.0	0.0	-0.2	-0.3	-0.4	-0.4		-0.3	-0.3	
of which: contribution from real GDP growth	-0.7	-0.9	-0.9			-0.5	0.1	-0.4	-1.8	-2.6	-3.0		-2.0	0.0	
Contribution from real exchange rate depreciation	-0.8	-0.6	0.1			-0.2	-0.5	-0.1	-0.2	-0.3	-0.4		...	...	
Other identified debt-creating flows	-112.2	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-112.2	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	115.2	-2.9	-0.2			3.5	5.7	-4.2	0.2	0.6	0.1		0.5	1.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	7.4			10.0	15.1	17.0	19.1	21.1	21.7		19.8	17.0	
of which: foreign-currency denominated	...	...	7.4			10.0	15.1	17.0	18.7	20.3	21.0		19.0	16.2	
of which: external	...	...	5.9			7.4	13.9	16.4	18.7	20.3	20.9		19.0	16.2	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	0.9	3.4	2.3			1.3	5.3	10.3	6.8	4.9	5.4		2.8	1.4	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	24.6			36.4	51.4	69.3	78.5	87.0	88.8		84.1	62.5	
PV of public sector debt-to-revenue ratio (in percent)	...	...	26.8			42.2	72.2	77.6	85.5	94.8	96.7		91.9	67.5	
of which: external 3/	...	...	21.4			31.4	66.5	74.9	83.5	91.1	93.1		88.2	64.3	
Debt service-to-revenue and grants ratio (in percent) 4/	1.6	0.4	3.0			1.3	2.4	4.5	3.8	1.9	5.4		7.1	5.6	
Debt service-to-revenue ratio (in percent) 4/	1.8	0.5	3.2			1.4	3.3	5.0	4.1	2.1	5.8		7.7	6.0	
Primary deficit that stabilizes the debt-to-GDP ratio	-1.5	4.6	1.0			-2.9	-5.2	4.9	2.2	2.7	3.7		2.1	-0.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.8	7.9	8.5	7.3	1.9	4.4	-0.4	1.8	6.4	8.5	9.4	5.0	6.1	-0.1	4.9
Average nominal interest rate on forex debt (in percent)	0.5	0.2	1.4	0.5	0.5	1.4	1.4	1.0	0.9	0.9	1.0	1.1	1.1	1.1	1.1
Average real interest rate on domestic debt (in percent)	-8.2	-5.7	-6.6	-5.9	4.4	...	...	...	...	6.4	2.1	4.2	1.0	-1.0	1.1
Real exchange rate depreciation (in percent, + indicates depreciation	-8.2	-5.6	0.6	-3.9	4.2	-1.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	12.2	8.7	8.0	9.6	3.7	10.0	7.2	6.1	7.5	7.8	7.4	7.7	6.4	8.4	6.5
Growth of real primary spending (deflated by GDP deflator, in percer	24.4	24.5	9.1	6.6	10.5	-5.6	19.3	1.2	-4.7	2.8	9.1	3.7	2.9	4.1	4.6
Grant element of new external borrowing (in percent)	...	...	...	...	...	47.9	43.1	53.2	48.4	38.1	39.2	45.0	48.1	48.1	...

Sources: Country authorities; and staff estimates and projections.

1/ It includes the gross debt contracted or guaranteed by the central government, excluding borrowing from the Central Bank of Liberia.

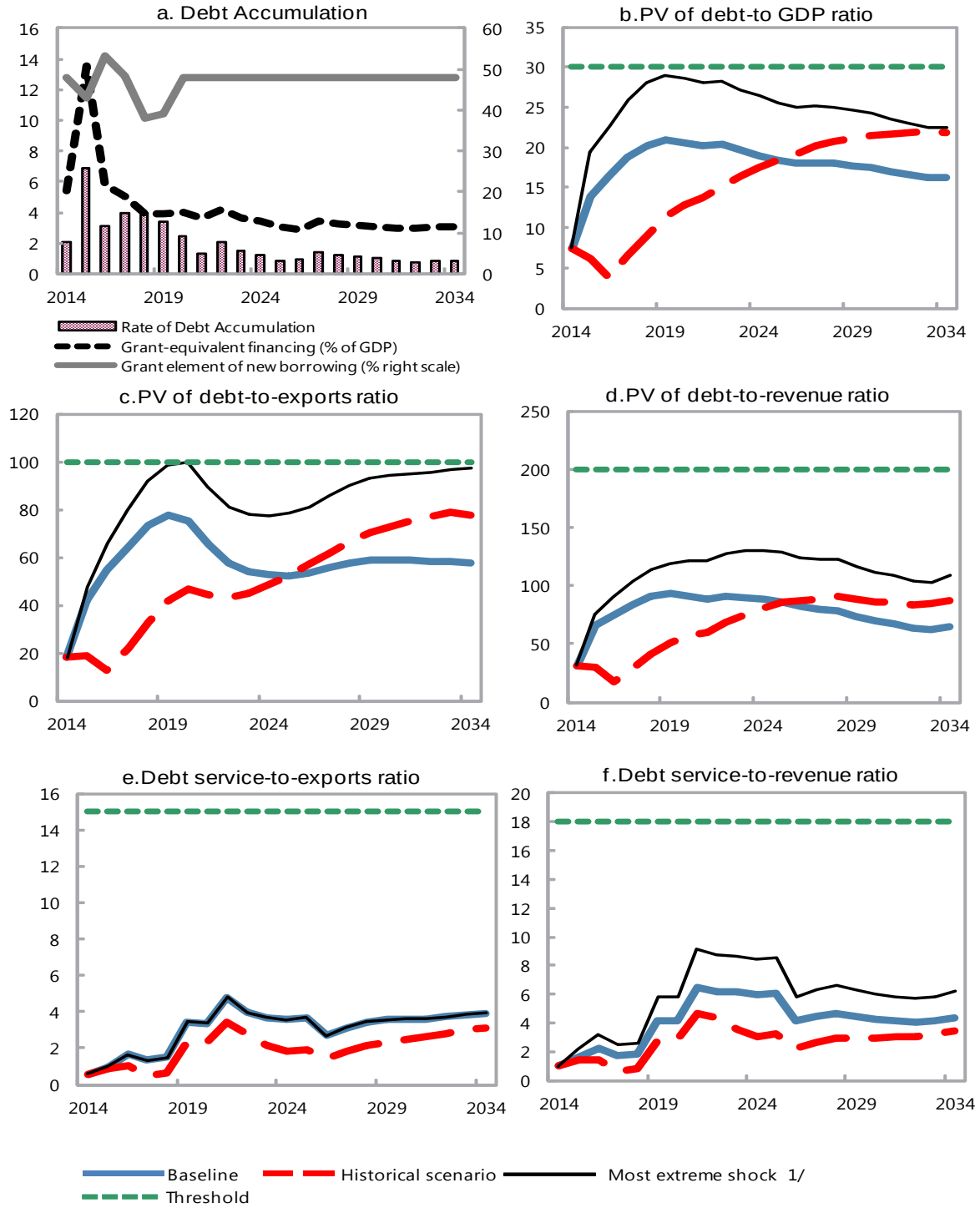
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

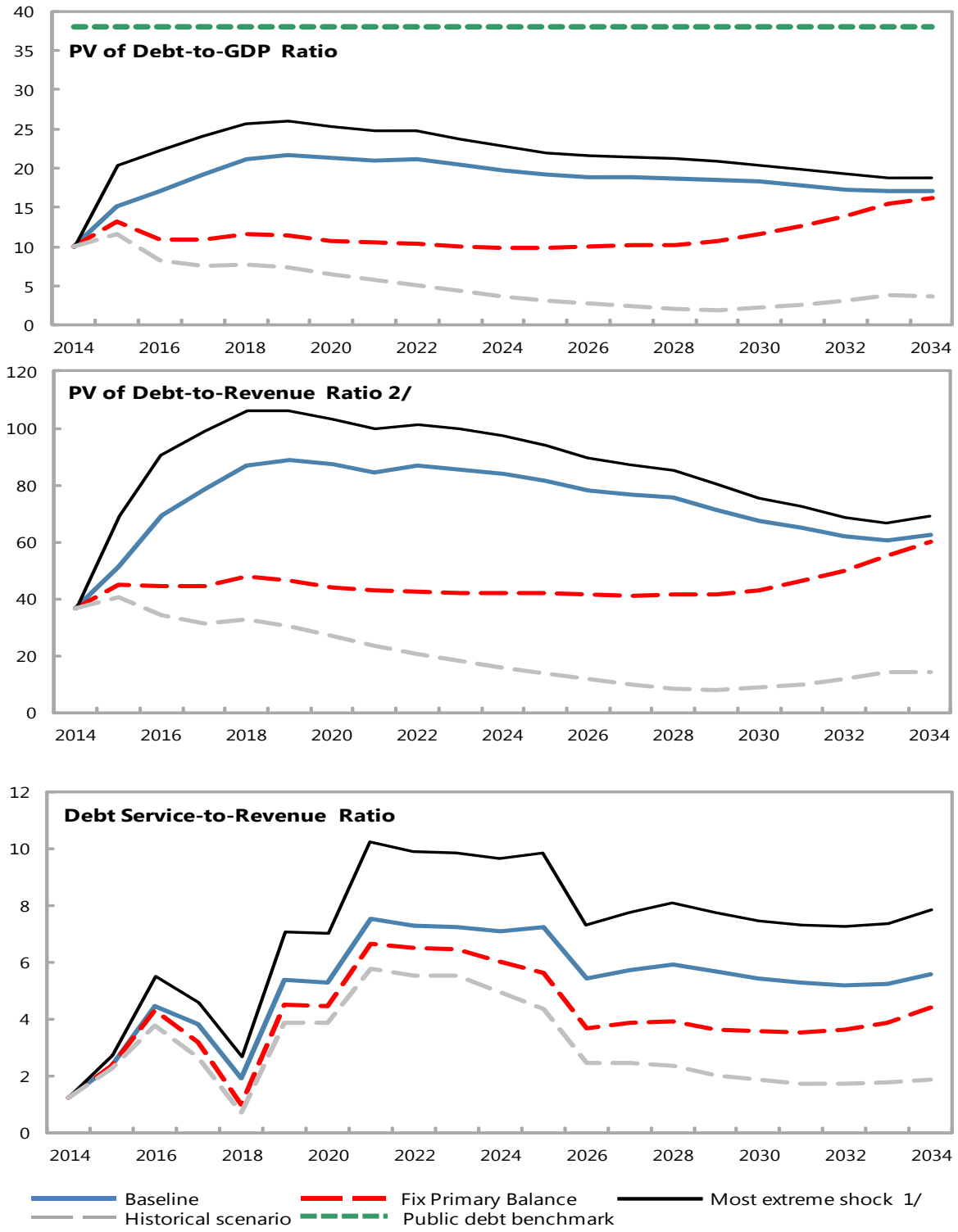
Appendix Figure 1. Liberia: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2014–34 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a Terms shock; in e. to a Growth shock and in figure f. to a One-time depreciation shock. In the alternative scenarios of external DSA, historical average and standard deviation of major variables are calculated by using the data from 2008 due to the structual change of the economy.

Appendix Figure 2. Liberia: Indicators of Public Debt Under Alternative Scenarios, 2014–34[1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2024.
2/ Revenues are defined inclusive of grants.